The Extremal Principle — problem 3.11

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Of n points in space, no four lie in a plane. Some of the points are connected by lines. We get a graph G with k edges.

(a) If G does not contain a triangle, then $k \leq \lfloor n^2/4 \rfloor$.

(b) If G does not contain a tetrahedron, then $k \leq \lfloor n^2/3 \rfloor$.